AI Solar Thermal Cooking System

Complete Component List and System Workflow

Project Documentation | June 2026

Project Summary

19	4	5	11
TOTAL COMPONENTS	ARDUINO SKETCHES	LIBRARIES USED	ARDUINO PINS USED

1. Bill of Materials (BOM)

#	COMPONENT	QTY	CATEGORY	CONNECTION / NOTES
1	Solar Panel 6V	2	Power	Connected in series to produce 12V; feeds charger module
2	18650 Battery Charging Module	1	Power	Input: ~12V from solar panels; charges 18650 battery with protection
3	18650 Li-ion Battery (3.7V)	1	Power	Energy storage for cloudy/night operation
4	LM2596 DC-DC Buck Converter	1	Power	Input: 3.7–12V; Output: regulated 5V for entire control circuit
5	Arduino Uno (ATmega328P)	1	Controller	Powered via 5V pin from LM2596
6	DS18B20 Waterproof Temp Sensor	1	Sensor	Pin D2 (1-Wire); measures water temperature; needs 4.7k pull-up
7	DHT11 Temp/Humidity Sensor	1	Sensor	Pin D3; measures ambient temperature and humidity
8	LDR (Light Dependent Resistor)	4	Sensor	Pins A1, A2, A3, A6; cross pattern for dual-axis sun tracking
9	Water Level Sensor	1	Sensor	Pin A0; analog reading, thresholds: LOW <200, FULL ≥300
10	SG90 Servo Motor (Pan)	1	Actuator	Pin D9 (PWM); horizontal sun tracking, range 45-135 degrees
11	SG90 Servo Motor (Tilt)	1	Actuator	Pin D10 (PWM); vertical sun tracking, range 45-135 degrees
12	5V Relay Module (single-channel)	1	Actuator	Pin D4; active LOW; controls water pump ON/OFF
13	DC Submersible Water Pump	1	Actuator	Connected via relay; circulates water through solar collector
14	I2C LCD 16x2 (PCF8574 backpack)	1	Display	Pins A4 (SDA), A5 (SCL); address 0x27 (auto-detected)
15	4.7k ohm Resistor	1	Passive	Pull-up resistor for DS18B20 DATA line to 5V
16	1k ohm Resistor	4	Passive	Voltage divider resistors for each LDR (to GND)
17	12V DC Power Supply (Adapter/Battery)	1	Heating	Dedicated supply for heating element only; common GND with Arduino
18	ON/OFF Switch (Toggle/MOSFET)	1	Heating	Series with +12V line; manual heater control
19	12V Heating Element (Nichrome/Ceramic)	1	Heating	Immersed in water tank; supplemental heating

2. Component Details by Category

2.1 Power Supply Components (5 items)

COMPONENT	SPECIFICATION	FUNCTION
Solar Panel 1 (6V)	6V photovoltaic panel	Connected in series with Panel 2 to produce combined 12V output
Solar Panel 2 (6V)	6V photovoltaic panel	Series with Panel 1; combined output feeds battery charger
18650 Charging Module	Input: ~12V; built-in charge controller	Charges 18650 battery with overcharge/over-discharge protection
18650 Li-ion Battery	3.7V nominal, rechargeable	Energy storage for operation during cloudy weather or night
LM2596 Buck Converter	Input: 3.7-12V; Output: 5V regulated	Steps down voltage to regulated 5V for Arduino and all peripherals

2.2 Controller (1 item)

COMPONENT	SPECIFICATION	FUNCTION
Arduino Uno	ATmega328P, 16 MHz, 14 digital + 6 analog pins	Central controller running sun tracking, sensor reading, pump control, and LCD display

2.3 Sensors (4 types, 7 units)

COMPONENT	QTY	PIN(S)	PROTOCOL	FUNCTION
DS18B20	1	D2	1-Wire	Waterproof temp sensor; range -55 to +125 C; 12-bit resolution; needs 4.7k pull-up
DHT11	1	D3	Digital	Ambient temperature (0-50 C) and humidity (20-80% RH)
LDR	4	A1, A2, A3, A6	Analog	Light sensors in cross pattern with 1k voltage dividers for dual-axis sun tracking
Water Level Sensor	1	A0	Analog	Detects water level; LOW <200, OK 200-299, FULL ≥300

2.4 Actuators (3 types, 4 units)

COMPONENT	QTY	PIN	SIGNAL	FUNCTION
Pan Servo (SG90)	1	D9	PWM	Horizontal sun tracking; range 45-135 degrees; 1-degree steps
Tilt Servo (SG90)	1	D10	PWM	Vertical sun tracking; range 45-135 degrees; 1-degree steps
Relay Module + Water Pump	1+1	D4	Digital (Active LOW)	Relay controls DC submersible pump for water circulation

2.5 Display (1 item)

COMPONENT	PIN(S)	PROTOCOL	FUNCTION
I2C LCD 16x2 (PCF8574)	A4 (SDA), A5 (SCL)	I2C, address 0x27	Alternating display: Page 1 = Water Temp + Level; Page 2 = Ambient + Servo Angles

2.6 Heating Circuit — Separate 12V Supply (3 items)

COMPONENT	SPECIFICATION	FUNCTION
12V DC Power Supply	Dedicated adapter or battery	Powers heating element only; separate from solar/LM2596 circuit; common GND with Arduino
ON/OFF Switch	Toggle switch or MOSFET	Manual control in series with +12V line to heater
12V Heating Element	Nichrome or ceramic, 12V DC	Immersed in water tank for supplemental heating when solar is insufficient

2.7 Passive Components (5 resistors)

COMPONENT	QTY	FUNCTION
4.7k ohm Resistor	1	Pull-up for DS18B20 1-Wire data bus (between D2 and 5V)
1k ohm Resistor	4	Voltage divider for each LDR (between analog midpoint and GND)

3. Arduino Pin Mapping

PIN	COMPONENT	SIGNAL TYPE	NOTES
D2	DS18B20 Water Temp Sensor	1-Wire Data	4.7k pull-up to 5V required
D3	DHT11 Ambient Temp/Humidity	Digital Data	—
D4	Relay Module (Water Pump)	Digital OUT	Active LOW
D9	Pan Servo (SG90)	PWM	Horizontal tracking, 45-135 degrees
D10	Tilt Servo (SG90)	PWM	Vertical tracking, 45-135 degrees
A0	Water Level Sensor	Analog IN	Threshold: LOW <200, FULL ≥300
A1	LDR — Top Left	Analog IN	1k voltage divider to GND
A2	LDR — Top Right	Analog IN	1k voltage divider to GND
A3	LDR — Bottom Left	Analog IN	1k voltage divider to GND
A4	LCD — SDA (I2C)	I2C Data	PCF8574 backpack, address 0x27
A5	LCD — SCL (I2C)	I2C Clock	—
A6	LDR — Bottom Right	Analog IN	1k voltage divider to GND
5V	All sensors, LCD, relay, servos, LDRs	Power	From LM2596 regulated output
GND	All components	Ground	Common GND shared with 12V heating circuit

4. Solar Power Chain

STAGE	COMPONENT	INPUT	OUTPUT	NOTES
1	Solar Panel 1 (6V)	Sunlight	+6V	In series with Panel 2
2	Solar Panel 2 (6V)	Sunlight	Combined +12V	(-) of Panel 1 connects to (+) of Panel 2
3	18650 Charging Module	~12V from panels	Regulated charge	Overcharge/discharge protection
4	18650 Li-ion Battery	From charger	3.7V nominal	Energy storage for night/cloudy conditions
5	LM2596 Buck Converter	3.7-12V	5V regulated	Adjustable via potentiometer
6	Arduino Uno (5V pin)	5V from LM2596	—	Powers entire control circuit

5. Software and Libraries

5.1 Arduino Libraries

LIBRARY	AUTHOR	PURPOSE
Servo.h	Built-in (Arduino)	PWM control for pan and tilt servo motors
Wire.h	Built-in (Arduino)	I2C communication for LCD display
DHT.h	Adafruit	Read temperature and humidity from DHT11
OneWire.h	Paul Stoffregen	1-Wire protocol for DS18B20 communication
DallasTemperature.h	Miles Burton	High-level interface for DS18B20 temperature readings

5.2 Arduino Sketches

#	FILE	PURPOSE	DESCRIPTION
1	solar_thermal_system.ino	Main System	Full integrated system — sun tracking, sensors, pump control, LCD display
2	solar_no_pump.ino	No-Pump Variant	Same as main but without water pump relay control
3	sensor_test.ino	Sensor Testing	Tests LCD, DS18B20, DHT11, and water level sensor with Serial output
4	pump_test.ino	Pump Testing	Simple relay ON/OFF test (5s cycles) to verify pump wiring

6. System Workflow

Step 1: Solar Energy Harvesting

Two 6V solar panels connected in series produce a combined 12V DC output. This feeds into the 18650 battery charging module which charges the Li-ion battery with overcharge/discharge protection.

Components: Solar Panel 1, Solar Panel 2, 18650 Charging Module, 18650 Battery

Step 2: Voltage Regulation

The LM2596 buck converter steps down the battery/charger output (3.7-12V) to a regulated 5V DC. This stable 5V powers the Arduino Uno and all connected sensors, servos, relay, and LCD display.

Components: LM2596 Buck Converter, Arduino Uno

Step 3: Dual-Axis Sun Tracking (every 100ms)

Four LDRs arranged in a cross pattern (Top-Left, Top-Right, Bottom-Left, Bottom-Right) detect sunlight direction. The Arduino calculates error signals:

Horizontal Error = (TL + BL) - (TR + BR) → adjusts Pan Servo on D9

Vertical Error = (TL + TR) - (BL + BR) → adjusts Tilt Servo on D10

If error exceeds the tolerance threshold (50 units), the corresponding servo moves by 1 degree. In darkness (all LDRs below 80), servos return to 90 degree home position.

Components: 4x LDR, 4x 1k Resistors, Pan Servo (SG90), Tilt Servo (SG90)

Step 4: Water Temperature Monitoring (every 1 second)

The DS18B20 waterproof sensor measures water temperature inside the cooking vessel at 12-bit resolution. Temperature readings are displayed on the LCD and can trigger safety logic.

Components: DS18B20, 4.7k Pull-up Resistor

Step 5: Ambient Monitoring (every 2 seconds)

The DHT11 sensor reads ambient air temperature and humidity around the cooking system. This data helps assess environmental conditions and system efficiency.

Components: DHT11

Step 6: Water Level Check (continuous)

An analog water level sensor on A0 continuously checks the water level. Status is categorized as: LOW (below 200) — needs refilling; OK (200 to 299) — normal; FULL (300 and above) — tank full.

Components: Water Level Sensor

Step 7: Water Circulation (pump control)

The relay module (Active LOW on D4) controls the DC submersible water pump. The pump circulates water through the solar thermal collector to maximize heat absorption. Pump test mode runs 5-second ON / 5-second OFF duty cycles.

Components: Relay Module, Water Pump

Step 8: LCD Display Output (every 3 seconds, alternating pages)

The I2C LCD 16x2 alternates between two information pages every 3 seconds:

Page 1: Water Temperature (degrees C) + Water Level Status (LOW/OK/FULL)

Page 2: Ambient Temperature, Humidity (%), Pan Servo Angle, Tilt Servo Angle

Components: I2C LCD 16x2 (PCF8574)

Step 9: Supplemental Heating (separate 12V circuit)

When solar energy alone is insufficient, a 12V heating element (nichrome/ceramic) immersed in the water tank provides supplemental heating. It is powered by a dedicated 12V DC power supply (completely separate from the solar circuit), controlled via a manual ON/OFF switch. The GND is shared (common) with the Arduino circuit.

Components: 12V DC PSU, ON/OFF Switch, Heating Element

7. Task Timing Summary

TASK	INTERVAL	FUNCTION NAME	COMPONENTS INVOLVED
Sun Tracking	100 ms	trackSun()	4x LDR → Pan and Tilt Servos
Water Temperature	1,000 ms	readWaterTemp()	DS18B20
Ambient Reading	2,000 ms	readAmbient()	DHT11
LCD Page Flip	3,000 ms	flipLCD()	LCD 16x2

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